A Documentation of TRMS Classroom Implementation for Experiment 1 and 2, and An Evaluation of the TRMS Approach

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This supplementary document provides classroom implementation guidance for the two thermoelectric real-time measurement system (TRMS) activities reported in the paper: experiment 1, temperature-gradient measurement and monitoring, and experiment 2, electrical characterization and extended logging. It also documents the pedagogical rationale of the approach and the forms of evidence used to evaluate classroom implementation. The uploaded supplementary draft already structures both experiments as guided-inquiry teaching plans for upper secondary and introductory undergraduate levels, with defined learning objectives, staged instructional procedures, and evidence of learning linked to Figures S1-S12.

The purpose of this document is to support teachers and readers who may wish to reproduce or adapt the TRMS for classroom use. It provides a structured teaching plan for both experiments, showing how the apparatus, software, and smartphone dashboard can be integrated into guided inquiry activities. In addition to the experimental configuration itself, the document describes intended learning outcomes, the sequence of classroom activities, and the forms of student evidence that may be collected during implementation. The source document frames both experiments within a guided-inquiry model and specifies equipment, core concepts, cognitive/behavioral/affective objectives, and staged procedures for conceptualization, instrumentation, results analysis, and evaluation.

Teaching Plan for Experiment 1:
Temperature-gradient measurement and monitoring

This activity is intended for upper secondary or introductory undergraduate physics and is designed for approximately 200 minutes of instruction. The activity is structured as guided inquiry, in which the teacher provides the problem context, conceptual support, and procedural scaffolding, while students construct the system, collect data, and interpret the results. The equipment includes a NodeMCU ESP8266, two MAX6675 modules, two thermocouples, a thermoelectric module, a battery, and the Blynk platform. The core concepts are temperature gradient (ΔT), thermoelectric phenomena, real-time temperature measurement, and IoT-based data monitoring.

1) Learning objectives:

The activity is designed to help students:

- Explain how a temperature difference is generated in a thermoelectric system and interpret the relationship between T_1, T_2, and ΔT
- Construct and operate a real-time temperature measurement system using MAX6675 modules and an ESP8266 microcontroller
- Appreciate the importance of real-time data acquisition and measurement accuracy in experimental physics.

2) Instructional sequence:

The implementation is organized into four stages.

2.1) Conceptualization (40 min)

The teacher introduces the guiding question, *"How is a temperature difference* (ΔT) *generated in a thermoelectric system?"* Students discuss heat transfer and thermoelectric effects, then receive background explanation on the Seebeck effect, temperature-gradient formation, thermocouple measurement, and the operation of the MAX6675 and NodeMCU. Evidence of this stage is associated with figures S1-S2.

2.2) Instrumentation (120 min)

Students assemble the circuit from a predefined schematic, integrating the NodeMCU, the two MAX6675 modules, thermocouples, the thermoelectric device, and the power source. They then verify wiring, initialize sensors, and establish communication with Blynk. The teacher supervises setup and troubleshooting. Evidence is linked to figures S3-S4.

2.3) Evaluation (40 min)

Students communicate their results through graphs, tables, or real-time dashboard displays, highlighting the measured temperatures and temperature difference. Evidence is linked to figure S5. In addition, Students explain the relationship between the measured temperatures and the behaviours of the thermoelectric system and reflect on data reliability and the effectiveness of the measurement system. Evidence is linked to figure S6.

Table S1.

Instructional Stage	Guided Inquiry Step	Learning Activities	Teacher Role	Student Role	Evidence of Learning
Conceptualization (40 min)	Problem/Question (10 min)	The teacher introduces the central guiding question: *"How is a temperature difference (ΔT) generated in a thermoelectric system?"* Students are prompted to consider the physical mechanisms underlying heat transfer and thermoelectric effects, and to relate these concepts to real-time measurement scenarios. This stage aims to activate prior knowledge and frame the investigation within a meaningful scientific context.	Facilitate inquiry and contextualization	Share prior knowledge and ideas	Figure S1
	Theory/Background (30 min)	The teacher provides the necessary theoretical foundation, including: (i) the principles of thermoelectric phenomena (e.g., Seebeck effects), (ii) the formation of a temperature gradient (*ΔT*) across a thermoelectric module, (iii) temperature measurement using thermocouples, and (iv) the operating principles of the MAX6675 thermocouple interface and the NodeMCU ESP8266 microcontroller. The explanation emphasizes the relationship between physical concepts and measurement instrumentation.	Provide conceptual explanation and scaffolding	Engage in questioning and conceptual understanding	Figure S2
Instrumentation (120 min)	Procedures/Design (60 min)	The teacher provides structured, step-by-step instructions for constructing the experimental system. Students assemble the circuit according to a predefined schematic, integrating the NodeMCU ESP8266, two MAX6675 modules, thermocouples, a thermoelectric device, and a power source (battery). The teacher also guides	Demonstrate, supervise, and ensure correct setup	Assemble system, upload code, troubleshoot	Figure S3

Instructional Stage	Guided Inquiry Step	Learning Activities	Teacher Role	Student Role	Evidence of Learning
		students in configuring the system, including wiring validation, sensor initialization, and establishing communication with the Blynk platform. Emphasis is placed on correct instrumentation practices and system reliability.			
	Results Analysis (60 min)	Students analyze the measured temperature data obtained from both thermocouples, identifying the hot-side (T_1) and cold-side (T_2) temperatures. They calculate the temperature difference $\Delta T = T_1 - T_2$ and interpret the results in relation to the experimental conditions. Students are encouraged to examine trends, identify possible sources of error, and relate the observed behaviours to thermoelectric principles.	Guide analytical reasoning	Perform calculations and interpretation	Figure S4
Evaluation (40 min)	Results Communication (20 min)	Students communicate their findings using appropriate scientific formats, such as graphs, tables, or real-time visualizations via the Blynk dashboard. They are encouraged to present data clearly, highlight key patterns, and explain the significance of the temperature difference within the system.	Facilitate presentation and discussion	Present findings and explain patterns	Figure S5
	Conclusions (20 min)	Students draw conclusions based on their analysis, explaining the relationship between the temperature difference (ΔT) and the behaviours of the thermoelectric system. They reflect on the effectiveness of the measurement system, the reliability of the data, and the underlying physical principles. This stage reinforces conceptual understanding and links experimental results to scientific reasoning.	Guide synthesis and conceptual linkage	Conclude and reflect	Figure S6

Figure S1. Verbal responses and discussion

Figure S2. Notes and responses

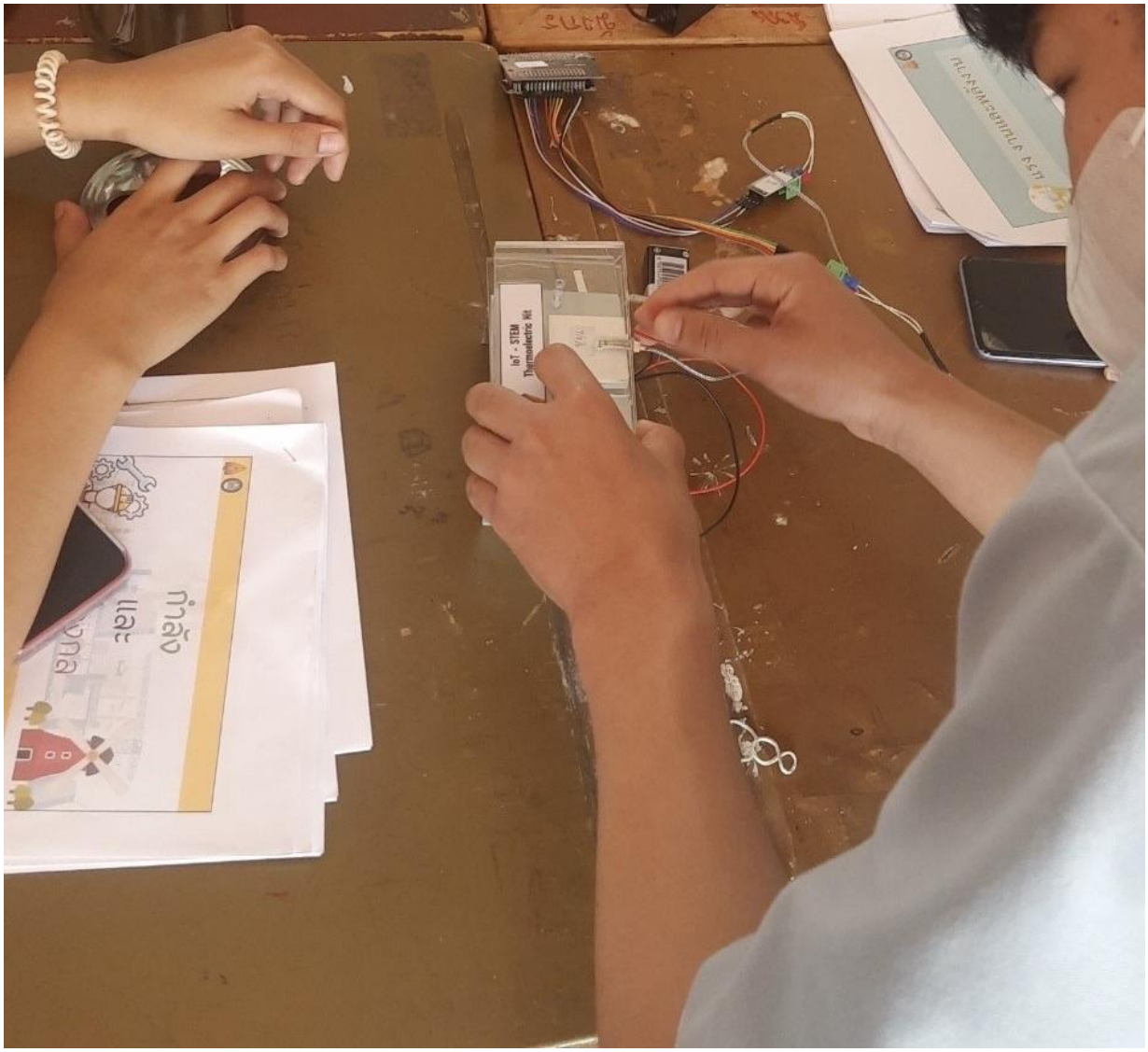

Figure S3. Functional measurement system

Figure S4. Data tables, calculations

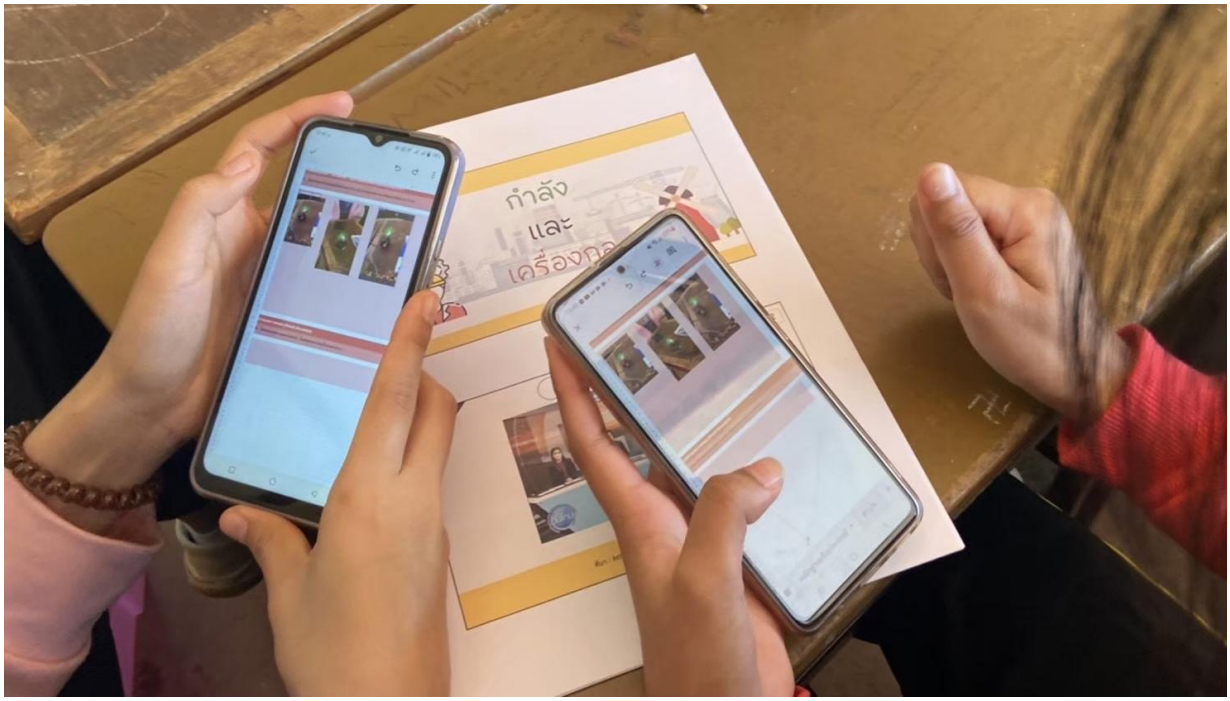

Figure S5. Graphs and dashboard outputs

Figure S6. Written and verbal conclusions

Teaching Plan for Experiment 2:
Electrical characterization and extended logging

This activity is also intended for upper secondary or introductory undergraduate physics and is designed for approximately 200 minutes. As with experiment 1, the lesson is structured as guided inquiry, with the teacher scaffolding the investigation while students assemble, test, and interpret a real-time thermoelectric measurement system. The equipment includes a WeMos D1 mini ESP8266, PZEM-017, MAX485, two MAX6675 modules, thermocouples, a thermoelectric module, a shunt resistor, a motor load, and the Blynk platform. The core concepts are thermoelectric energy conversion, the relationship between ΔT and electrical output, and real-time electrical measurement.

1) Learning objectives:

The activity is designed to help students:

- Explain thermoelectric energy conversion and interpret the relationship between $\Delta T$and electrical output, including voltage, current, power, and energy.
- Construct and operate an integrated measurement system using the thermoelectric module, PZEM-017, and microcontroller-based instrumentation.
- Appreciate the relevance of thermoelectric systems in energy-conversion applications.

2) Instructional sequence:

The implementation is organized into four stages.

2.1) Conceptualization (40 min)

The teacher introduces the guiding question, *"How can thermoelectric systems generate usable electrical energy?"* Students discuss how temperature difference, electrical output, and load behaviours are related. The teacher then explains thermoelectric energy conversion, the Seebeck effect, the relation between $\Delta T$and electrical output, the electrical quantities V, I, $P = VI$, and $\Delta E = P\Delta t$, and the role of the PZEM-017, shunt resistor, thermocouples, MAX6675 modules, and WeMos D1 mini. Evidence is linked to figures S7-S8.

2.2) Instrumentation (120 min)

Students assemble the full experimental configuration, integrating the WeMos D1 mini, PZEM-017, MAX485 RS485 interface, two MAX6675 modules, thermocouples, thermoelectric module, shunt resistor, and motor load. The teacher supervises wiring, power supply arrangement, sensor initialization, and communication with Blynk, with attention to electrical safety and stable operation. Evidence is linked to figures S9-S10.

2.3) Evaluation (40 min)

Students communicate their findings using graphs, tables, and real-time dashboard displays, especially trends in electrical parameters. Evidence is linked to figure S11. Students conclude how temperature difference, electrical output, and energy-conversion behaviours are related, including conversion to mechanical output through the motor load where applicable. They also reflect on instrumentation reliability and broader implications for thermoelectric energy conversion. Evidence is linked to figure S12.

Table S2.

Instructional Stage	Guided Inquiry Step	Learning Activities	Teacher Role	Student Role	Evidence of Learning
Conceptualization (40 min)	Problem/Question (10 min)	The teacher introduces the guiding question: *"How can thermoelectric systems generate usable electrical energy?"* This question directs students to consider thermoelectric energy conversion as a process that links temperature difference, electrical output, and load behavior. Students are encouraged to consider how thermal conditions can be converted into measurable electrical quantities such as voltage, current, power, and energy.	Initiate inquiry	Discuss and hypothesis	Figure S7
	Theory/Background (30 min)	The teacher provides the conceptual foundation necessary for the investigation, including: (i) thermoelectric energy conversion and the Seebeck effect, (ii) the relationship between temperature difference (ΔT) and electrical output, (iii) electrical parameters including voltage (V), current (I), power ($P=VI$), and energy ($\Delta E=P\Delta t$), (iv) the role of the thermoelectric module as an energy conversion device, (v) the function of the PZEM-017 power measurement module and shunt resistor, and (vi) the principles of temperature measurement using thermocouples, MAX6675 modules, and the WeMos D1 mini ESP8266 microcontroller. The explanation emphasizes the connection between physical principles, measurement instrumentation, and real-time monitoring.	Provide theoretical framework	Engage in understanding	Figure S8

Instructional Stage	Guided Inquiry Step	Learning Activities	Teacher Role	Student Role	Evidence of Learning
Instrumentation (120 min)	Procedures/Design (60 min)	The teacher provides structured step-by-step procedures for circuit construction and system setup. Students assemble the experimental configuration according to the predefined schematic, integrating the WeMos D1 mini, PZEM-017, MAX485 RS485 interface, two MAX6675 modules, thermocouples, a thermoelectric module, a shunt resistor, and a motor load. The teacher also guides students in system configuration, including verification of wiring connections, power supply arrangement, initialization of sensors, and communication with the Blynk platform. Emphasis is placed on correct instrumentation practices, electrical safety, and stable operation of the integrated measurement system	Demonstrate and supervise	Assemble and test system	Figure S9
	Results Analysis (60 min)	Students analyze the measured thermal and electrical data obtained during the experiment. They interpret the hot-side and cold-side temperatures, determine the temperature difference (ΔT), and examine the corresponding electrical parameters, including voltage, current, power, and energy. Students are encouraged to identify relationships between changes in temperature gradient and variations in electrical output, as well as to consider the influence of load conditions on system behaviours. The analysis also promotes reflection on possible sources of uncertainty and the reliability of the acquired data.	Guide analysis	Interpret data	Figure S10

Instructional Stage	Guided Inquiry Step	Learning Activities	Teacher Role	Student Role	Evidence of Learning
Evaluation (40 min)	Results Communication (20 min)	Students communicate their findings using scientific representations such as graphs, tables, or real-time data visualized on the Blynk dashboard. In particular, they present trends in electrical parameters	Facilitate communication	Present findings	Figure S11
	Conclusions (20 min)	Students conclude the relationship among temperature difference, electrical output, and energy-conversion behaviours in the thermoelectric system. They explain how the experimental data demonstrate the conversion of thermal energy into electrical energy, and, where applicable, into mechanical energy through the motor load. Students also reflect on the effectiveness of the measurement system, the reliability of the instrumentation, and the broader implications of thermoelectric energy conversion in the context of renewable and alternative energy technologies.	Guide synthesis	Summarize and reflect	Figure S12

Figure S7. Responses

Figure S8. Notes

Figure S9. Functional setup

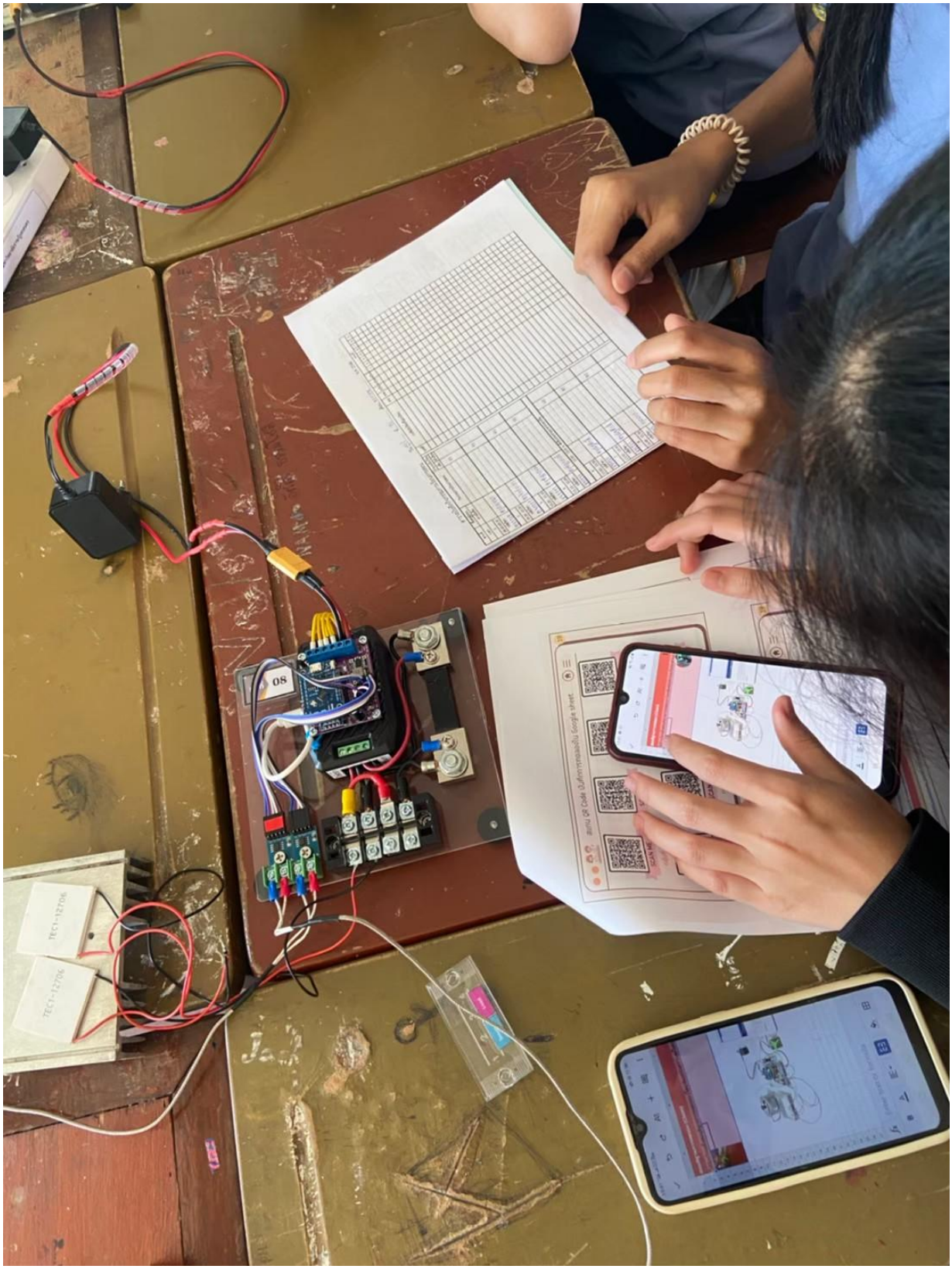

Figure S10. Data tables and graphs

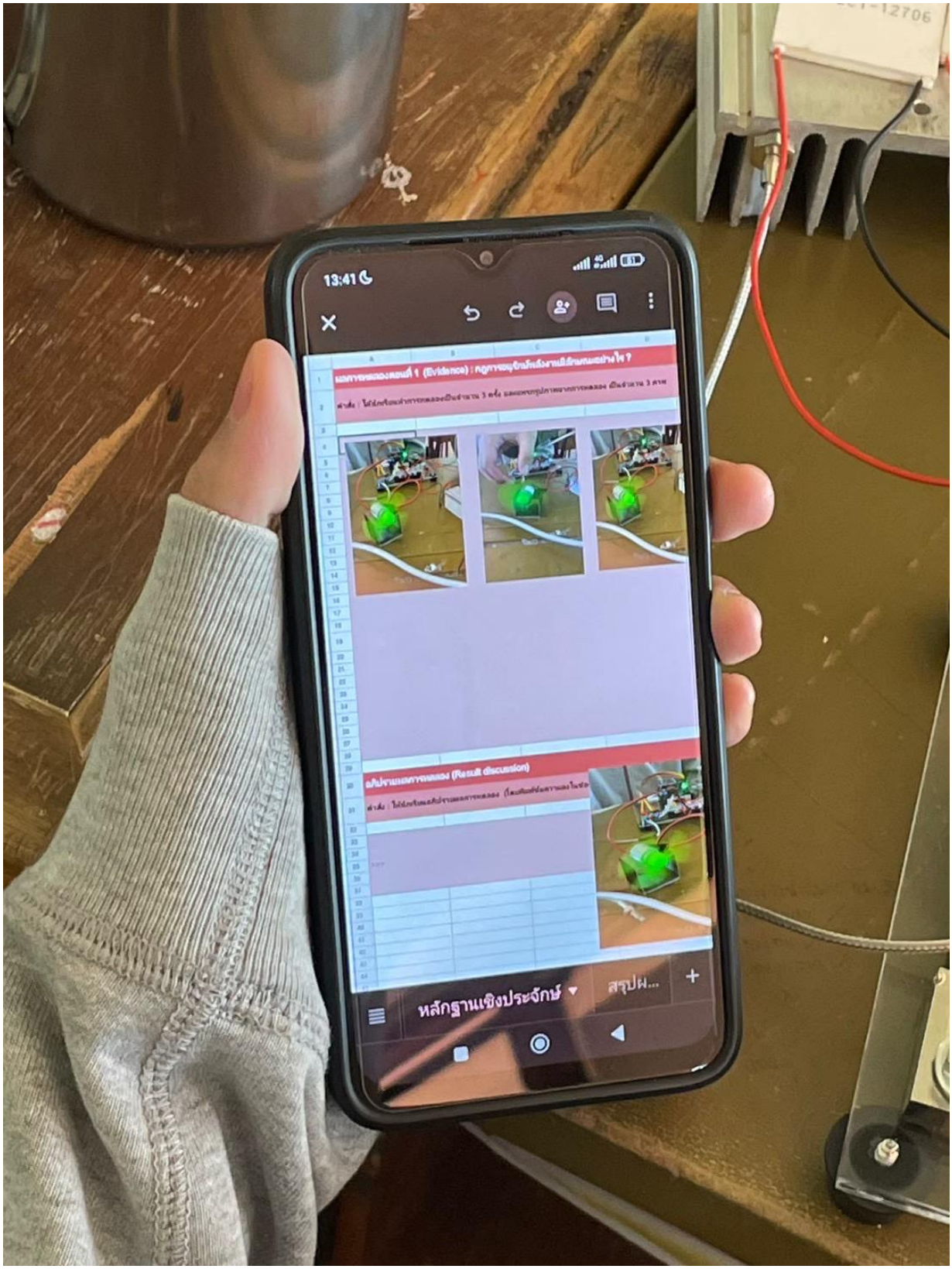

Figure S11. Visual outputs

Figure S12. Conclusions

Evaluation of the TRMS Approach

The TRMS learning activity was implemented with 40 students (n = 40) working in small groups. With a provided wiring diagram and a short starter program, groups were able to stream measurements to a smartphone dashboard and generate a stable current–time plot within the session. Initial difficulties were mainly practical (for example, loose breadboard connections, reversed polarity, and intermittent network connectivity) and were typically resolved quickly using a brief troubleshooting checklist. After the live plot was established, students readily explored "what-if" modifications (such as increasing heating, improving cooling, or changing load conditions) and used the immediate response of the trace to support explanations linking thermal conditions to electrical output.

A brief four-item pre-test and post-test questionnaire indicated positive shifts in renewable-energy awareness and attitudes. Students reporting environmental consciousness and familiarity with renewable energy technologies increased from 16 out of 40 (40%) to 34 out of 40 (85%). The proportion who strongly agreed that renewable energy projects in their region would positively contribute to awareness and understanding of benefits increased from 23 out of 40 (57.5%) to 29 out of 40 (72.5%). Agreement that the government should support pilot demonstration projects for every type of renewable energy technology increased from 35 out of 40 (87.5%) to 38 out of 40 (95%). Willingness to consider buying electrical energy from a renewable source even if slightly more expensive increased from 29 out of 40 (72.5%) to 34 out of 40 (85%).

Although these results are based on self-report measures, the pattern of gains across all four items suggests meaningful learning benefits beyond technical skill development. In particular, the approach appears to strengthen energy awareness by helping students connect a concrete, observable energy-conversion process to broader renewable-energy considerations. The real-time current trace made the thermal-to-electrical conversion visible and discussable during the activity, which likely supported students' increased familiarity with renewable technologies and their greater willingness to endorse local projects, government-supported demonstrations, and renewable electricity purchasing decisions.